



The Journal of Development Studies

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/fjds20>

Conditional Cash Transfers, Adult Work Incentives, and Poverty

Emmanuel Skoufias^a & Vincenzo Di Maro^b

^a The World Bank , USA

^b University College London , UK

Published online: 18 Sep 2008.

To cite this article: Emmanuel Skoufias & Vincenzo Di Maro (2008) Conditional Cash Transfers, Adult Work Incentives, and Poverty, The Journal of Development Studies, 44:7, 935-960, DOI: [10.1080/00220380802150730](https://doi.org/10.1080/00220380802150730)

To link to this article: <http://dx.doi.org/10.1080/00220380802150730>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly

Conditional Cash Transfers, Adult Work Incentives, and Poverty

EMMANUEL SKOUFIAS* & VINCENZO DI MARO**

*The World Bank, USA, **University College London, UK

Final version received August 2007

ABSTRACT *Conditional cash transfer (CCT) programmes aim to alleviate poverty through monetary and in-kind benefits, as well as reduce future incidence of poverty by encouraging investments in education, health and nutrition. The success of CCT programmes at reducing poverty depends on whether, and the extent to which, cash transfers affect adult work incentives. In this paper we examine whether the PROGRESA programme of Mexico affects adult participation in the labour market and overall adult leisure time, and we link these effects to the impact of the programme on poverty. Utilising the experimental design of PROGRESA's evaluation sample, we find that the programme does not have any significant effect on adult labour force participation and leisure time. Our findings on adult work incentives are reinforced further by the result that PROGRESA leads to a substantial reduction in poverty. The poverty reduction effects are stronger for the poverty gap and severity of poverty measures.*

I. Introduction

Mean-tested conditional cash transfer (CCT) programmes are increasingly popular in developing countries as a useful tool for poverty alleviation. Examples of such programmes include PROGRESA/*Oportunidades* in Mexico, *Bolsa Familia* in Brazil, *Bono de Desarrollo Humano* in Ecuador, *Familias en Accion* in Colombia, PRAF in Honduras, PATH in Jamaica, and *Red de Proteccion Social* (RPS) in Nicaragua, among others. Targeting their benefits directly to populations in extreme poverty, primarily in rural areas, such programmes aim to alleviate current poverty through monetary and in-kind benefits, as well as reduce future levels of poverty by encouraging investments in education, health and nutrition.

The success of CCT programmes at reducing current poverty depends on whether, and the extent to which, cash transfers affect adult work incentives. In all of the programmes mentioned above, once a household is selected as eligible for the programme, usually through geographic targeting or household level means-testing

Correspondence Address: Emmanuel Skoufias, The World Bank, 1818 H Street NW, Washington DC 20433-USA. Email: eskoufias@worldbank.org

or both, the level of benefits is not affected by the work decisions of the household members or the income level of the household. Thus, the main effect of CCT on the labour supply of adults may be a pure income effect.¹ This is in contrast to most welfare programmes in the US and Canada, that have explicit disincentives to work. For example, in the Aid to Families with Dependent Children (AFDC) in the US, the level of benefits is affected by work decisions as work income is effectively taxed by reducing the level of benefits provided.

The extent to which the transfers of CCT programmes result to significant income effects on adult leisure and consumption can only be determined empirically. The incentive effects of welfare programmes on labour supply have been the subject of intense scrutiny primarily in the US, the UK and Canada (see Stafford, 1985; Moffitt, 1992; Blundell and MaCurdy, 1999; Widerquist, 2005). In developing countries, however, the evidence regarding the labour supply responses of adults to transfer programmes is quite scarce. The study of Sahn and Alderman (1996), one of the rare studies in this topic, suggests that the labour supply effect of a rice subsidy programme in Sri Lanka is significantly large. Yet, in a recently published ex-ante micro simulation study of the impact of the Bolsa Escola programme on poverty in Brazil, the income effect of the transfers on adult labour supply is assumed to be negligible (Bourguignon et al., 2003). In addition to this, Leon et al. (2001) found a negative labour supply effect of an unconditional transfer programme.

Our paper sheds light on these issues using data from a large conditional cash transfer programme in the poor rural areas of Mexico, called PROGRESA (Health, Education and Nutrition Programme). With a cash transfer of 20 per cent of pre-programme consumption, PROGRESA has the potential of affecting adult work incentives of both programme participants as well as non-participants. For eligible households, the income effect of the cash transfer may be weakened by the direct and indirect time costs associated with adhering to the requirements of the programme. In addition, the means testing associated with CCT programmes may also affect the incentives of both eligible and non-eligible households. On the one hand, individuals who are not eligible for the programme's benefits may also have the incentive to work less or earn (or report) a lower income hoping to become eligible for the programme in future rounds of expansion of the programme. On the other hand, the possibility of future means-tests may impact on the labour supply and investment choices of currently eligible households.

The empirical analysis uses panel data from households surveyed for the purpose of evaluating the impact of PROGRESA on basic indicators of household investment in human capital.² A distinguishing feature of the PROGRESA data is that they are based on an experimental design, with randomisation of the coverage of the programme at the locality level. The empirical methodology consists of comparing conditional means of key outcome variables (such as labour market participation, hours of leisure, and poverty rates) between households living in villages covered by the PROGRESA programme (treatment villages) and households living in comparable villages that are out of the programme (controls). An additional advantage offered by the design of the sample, is that we can also examine the potential effects of the programme on the labour supply of the non-eligible households living in the treated communities (that is, the communities covered by PROGRESA).

We investigate three main questions. First, we examine whether being eligible or ineligible for PROGRESA benefits affects adult labour force participation. Second, we study the effects of PROGRESA on adult leisure hours. Leisure may be an important determinant of welfare and beneficiary households may choose to increase their welfare by using the cash transfers of the programme to 'buy' more leisure. Finally, we analyse the impact of PROGRESA on poverty measures based on household income. Poverty measures offer the advantage of being simple, albeit imperfect, summary measures of the effects of the programme on the communities with both eligible and ineligible households exposed to the programme. Significant labour supply response among households in communities covered by the programme may result in small impacts of the programme on poverty.

The organisation of the paper is as follows. Section II briefly describes the PROGRESA programme and the data used. Section III illustrates the econometric specification and estimation approach behind our results. Section IV shows and discusses the results regarding the impact of PROGRESA on adult labour force participation and leisure. The impact of PROGRESA on poverty is studied in section V, while section VI concludes.

II. A Brief Description of PROGRESA and of the Data Used

PROGRESA, initially implemented by the Mexican Federal government in 1997, adopts an integrated approach to combating the different causes of poverty by intervening simultaneously in the areas of health, education and nutrition. By 2004, the programme which was renamed *Oportunidades*, included nearly 5 million families in 72,345 localities in all 31 states. The total annual budget of the programme in 2004 was around US\$2.5 billion, or 0.3 per cent of the gross domestic product.

The education component of PROGRESA is designed to increase school enrollment among youth in Mexico's poor rural communities by making education grants available to pupils' mothers, who are then required to have their children attend school regularly. In localities where PROGRESA currently operates, households that have been characterised as poor, and have children enrolled in grades 3–9, are eligible to receive these educational grants every two months. The levels of these grants were determined by taking into account, among other factors, what a child would earn in the labour force or contribute to family production. The educational grants are slightly higher at the secondary level for girls, given their propensity to drop out at earlier ages. Every two months, confirmation of whether children of beneficiary families attend school more than 85 per cent of the time is submitted to PROGRESA by school teachers and directors, and this triggers the receipt of a bi-monthly cash transfer for school attendance.

In the area of health and nutrition, PROGRESA brings basic attention to health issues and promotes healthcare through free preventative interventions, such as nutritional supplements, and education on hygiene and nutrition as well as monetary transfers for the purchase of food. Receipt of monetary transfers and nutritional supplements are tied to mandatory healthcare visits to public clinics. This aspect of the programme emphasises targeting its benefits to children under five years, and pregnant and lactating women, and is administered by the Ministry of Health and by

IMSS-*Solidaridad*, a branch of the Mexican Social Security Institute, which provides benefits to uninsured individuals in rural areas.

Nutritional supplements are given to children between the ages of four months and two years, and to pregnant and breast-feeding women. If signs of malnutrition are detected in children between the ages of 2–5 years, nutritional supplements will also be administered. The nutritional status of beneficiaries is monitored by mandatory visits to the clinic and is more frequently monitored for children five years of age and under, pregnant women and lactating women. Upon each visit, young children and lactating women are measured for wasting (weight-for-height), stunting (height-for-age), and weight-for-age. An appointment monitoring system is set up and a nurse or doctor verifies adherence. The healthcare professionals submit every two months certification of beneficiary visits to PROGRESA, which triggers the receipt of the cash transfer for food support.

The average monthly payment, distributed every two months to the mothers in beneficiary families amounts to 20 per cent of the value of monthly consumption expenditures prior to the initiation of the programme.³ Working counter to this transfer are the direct and indirect costs associated with participation in the programme. For example, one of the conditions of the PROGRESA programme is that households eligible for the programme were required to stop receiving benefits from other pre-existing programmes such as *Niños de Solidaridad*, *Abasto Social de Leche, de Tortilla* and the National Institute of Indigenous people (INI).⁴ In addition to these direct costs, there are some indirect costs associated with complying with the programme's requirements. Such costs include the time costs of taking children to school and the health centre, waiting in line at the health centre, attending educational seminars on nutrition and hygiene, and travelling to the localities where payments are being distributed.

The data used in this paper consists of the sample of communities and households surveyed between November 1997 and November 1999, for the purposes of the evaluation of the PROGRESA programme. In order to obtain a credible evaluation of the potential impact of the programme, the PROGRESA administration decided to adopt an experimental design that allows one to compare households before and after the initiation of PROGRESA with similar households that were not yet covered by the programme. Specifically, the full sample used in the evaluation of PROGRESA consists of repeated observations (panel data) collected for 24,000 households from 506 localities in the seven states of Guerrero, Hidalgo, Michoacan, Puebla, Queretaro, San Luis Potosi and Veracruz. The opportunity of having a randomised design came from the fact the programme was introduced in phases, due to budgetary constraints. In fact, of the 506 localities, 320 localities were assigned to the treatment group (where PROGRESA was in operation) and 186 localities were assigned as controls.⁵ As originally planned, the localities serving the role of a control group started receiving PROGRESA benefits by December 1999.

The selection of beneficiaries into PROGRESA was a two-stage process (Skoufias et al., 2001). In the first stage, using national census data, poor communities with schooling and health infrastructure were identified. In the second stage, households within the selected poor communities were classified as eligible or ineligible based on the socioeconomic data collected by a census of all the households residing in the communities to be covered by the programme. On average, in the evaluation sample,

78 per cent of the households were classified as eligible for programme benefits. However, the fraction of households that actually ended up receiving the PROGRESA cash transfers during the two-year interval covered by the evaluation sample is just under 65 per cent, due to administrative errors and delays in the final registration of beneficiary households.⁶

The 1997 baseline household census called ENCASEH (*Encuesta de Características Socioeconómicas de los Hogares*) was followed by a number of socioeconomic household surveys – see *Encuesta de Evaluación de los Hogares* (ENCEL) – designed to collect information for the evaluation of PROGRESA. The first evaluation survey took place in March 1998, before the initiation of benefits in May 1998. The remaining surveys were conducted after beneficiary households in treatment villages started receiving benefits from PROGRESA. One round of surveys took place in November 1998, which was well after most households received some benefits as part of their participation in the programme. The next two waves took place in June 1999 and November 1999. A number of core questions about the demographic composition of households and their socioeconomic status were applied in each round of the survey.

Data used in this paper are drawn from four rounds. The first round (R1) took place in November 1997, the second round (R2) in November 1998, the third round (R3) in June 1999, and the fourth round (R4) took place in November 1999.

The November 1997 ENCASEH survey as well as the November 1998, June 1999, and November 1999 ENCEL surveys collected detailed information on income earned or received from a variety of sources for each individual in the household.⁷ The survey instrument used to collect individual and household income for these various sources changed significantly beginning with the November 1998 survey. With this caveat in mind, it should be noted that a serious effort was made to maintain comparability of income by source across the survey rounds. The various sources of income (excluding the PROGRESA cash transfers) were transformed into monthly income and then aggregated into four main sources of income: labour income; income from self employment (such as income from sewing, food preparation, construction or carpentry, commerce, produce transportation, repairs and laundry or cooking); other income (such as pensions, interest income, rents and community profits); and government transfers (such as educational scholarships from *Niños de Solidaridad*, benefits from INI, PROBECAT, *Empleo Temporal* and Procampo).

The income data by source are useful at quantifying the direct costs associated with participation in the PROGRESA programme. Children can contribute income to families by working for wages or by being recipients of cash transfers from other government transfer programmes excluding PROGRESA. Table 1 presents how the income contributed to families by children between ages 8–17 evolved across different survey rounds in the treatment and control villages (using only eligible households $E = 1$). Panel (a) in Table 5 reveals that the total (labour + other) income (excluding PROGRESA cash transfers) beneficiary families received from children had declined in both treatment and control localities since the initiation of PROGRESA in November 1998. The mean total income contributed by children in November 1998 is slightly lower among treatment villages compared to the control villages and the gap gets even bigger by the June 1999 round. By November 1999,

this gap is completely eliminated as control households are already incorporated into PROGRESA.⁸ Panels (b) and (c) in Table 1 break down total income from children into its two components, that is income from labour and other income that consists mainly of government transfers. These graphs reveal that the differences in mean total income from children in beneficiary households and eligible households in control localities are primarily due to drops in the child-related income beneficiary families received from other government programmes. It also appears that there are no significant differences in the labour income of children from beneficiary households in treatment localities and the labour income contribution of children in eligible households in control villages.

In addition to the potential income losses from children's work and benefits, households receiving PROGRESA benefits are also required to give up benefits from programmes such as *Abasto Social de Leche*, *de Tortilla* and the National Institute of Indigenous people (INI). Figure 1 makes clear that among beneficiary households (that is, those that received any PROGRESA benefits between May 1998 and November 1999 in the treatment villages), the incidence of benefits received from DIF, *Ninos de Solidaridad* and *Abasto Social de Leche* decreased dramatically. In combination, the preceding discussion suggests that the effect of PROGRESA on household income and poverty may not be adequately summarised by the size and incidence of the cash transfers.

III. Econometric Specification and Estimation

We begin with a brief discussion of the estimation approach that underlies our estimates of the impact of PROGRESA on labour force participation and leisure time.

The estimation of the impact of PROGRESA (for labour force participation and poverty rates) is based on the difference-in-differences estimator. This estimator is based on comparing differences between the treatment and control groups before and after the start of the PROGRESA programme.⁹ It offers the advantage that any

Table 1. Mean household income from children (Among beneficiary households with children aged 8–17)

	November 1997		November 1998		June 1999		November 1999	
	Treatment	Control	Treatment	Control	Treatment	Control	Treatment	Control
Total								
Mean	150	147	132	145	93	116	111	119
SD	137	135	112	198	99	115	147	138
Labour								
Mean	113	114	117	105	86	85	102	108
SD	140	140	130	201	112	117	151	137
Other								
Mean	37	34	15	40	7	31	9	11
SD	12	11	23	7	22	5	9	6

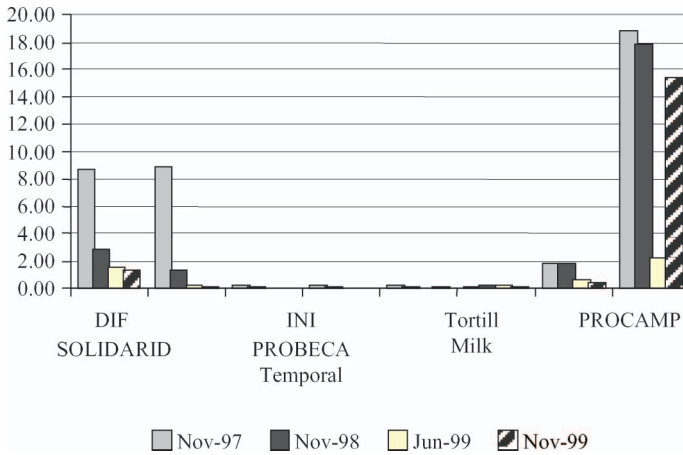


Figure 1. Percentages of households in treatment localities that receive the transfers from other programmes and PROCAMPA

pre-programme differences between the treatment and control group are eliminated in the estimation of impacts. Under the assumption that any unobserved heterogeneity between the treatment and control groups is fixed over time, the difference-in-differences estimator (2DIF, discussed in detail below) eliminates this heterogeneity. Specialised empirical specifications are implemented for the labour force participation and poverty estimation, and are discussed below. We also use a number of control variables which may be useful for reducing any remaining statistical bias.

To begin with, consider the case where there are data available for treatment and control households before and after the start of the programme. Restricting the sample to eligible households only ($E = 1$), the following regression equation defines a model that can nest various ‘difference’ estimators allowing for control of individual, household and locality observed characteristics:

$$Y(i, t) = \alpha + \beta_T T(i) + \beta_{R2}(R2) + \beta_{R3}(R3) + \beta_{R4}(R4) + \beta_{TR2}(T(i) \times R2) \\ + \beta_{TR3}(T(i) \times R3) + \beta_{TR4}(T(i) \times R4) + \sum_{j=1}^J \theta_j X_j(i, t) + \eta(i, t) \quad (1)$$

$Y(i, t)$ denotes the value of the outcome indicator of interest for household, individual or population i in period/round t , α , β , and θ are fixed parameters to be estimated, $T(i)$ is a binary variable taking the value of 1 if the household resides in a treatment community and 0 otherwise (that is, for control communities), $R2$, $R3$, and $R4$ are binary variables equal to 1 for the second, third and fourth rounds of the survey respectively, and equal to zero otherwise. Note that the first round is the baseline round prior to the initiation of the programme. The vector X summarises household (and possibly village) characteristics and η is an error term summarising the influence of random disturbances.

To better understand the preceding specification it is best to consider the case of only two rounds: one before the initiation of the programme and the other after the start of the programme (for example, round 2 or $R2 = 1$, 0 otherwise). One may then divide the parameters into two groups: one group summarising differences in the conditional mean of the outcome indicator before the start of the programme (that is, α , β_T) and another group summarising differences after the start of the programme (β_R , and β_{TR}). Specifically, the coefficient β_T allows the conditional mean of the outcome indicator to differ between eligible households in treatment and control localities before the initiation of the programme, whereas the remainder of the parameters allow the passage of time to have a different effect on households in treatment and control localities. For example, the combination of parameters β_R and β_{TR} allow the differences between eligible households in treatment and control localities to be different after the start of the programme.

Specifically, given the preceding specification, the conditional mean values of the outcome indicator for treatment and control groups before and after the start of the programme are as follows:

$$[E(Y|T = 1, R2 = 1, \mathbf{X})] = \alpha + \beta_T + \beta_R + \beta_{TR} + \sum_j \theta_j X_j \quad (2)$$

$$[E(Y|T = 1, R2 = 0, \mathbf{X})] = \alpha + \beta_T + \sum_j \theta_j X_j \quad (3)$$

$$[E(Y|T = 0, R2 = 1, \mathbf{X})] = \alpha + \beta_R + \sum_j \theta_j X_j \quad (4)$$

$$[E(Y|T = 0, R2 = 0, \mathbf{X})] = \alpha + \sum_j \theta_j X_j \quad (5)$$

The advantage offered by the 2DIF estimator is that it provides an estimate of the impact of the programme that is net of any pre-programme differences between treatment and control households and/or any time trends or aggregate effects in changes of the values of the outcome indicator. By comparing before and after differences between treatment and control households (or differences between treatment and control households after and before the programme), one is able to get an estimate of the impact of the programme (summarised by the single parameter β_{TR}).

$$\begin{aligned} 2DIF = & [E(Y|T = 1, R2 = 1, \mathbf{X}) - E(Y|T = 1, R2 = 0, \mathbf{X})] \\ & - [E(Y|T = 0, R2 = 1, \mathbf{X}) - E(Y|T = 0, R2 = 0, \mathbf{X})] \end{aligned} \quad (6)$$

Using the terminology of Heckman et al. (1999), the parameter β_{TR} provides an estimate of the ‘intent to treat effect,’ which is inclusive of the operational efficiency or inefficiency of the programme implementation. Thus, β_{TR} provides a lower bound estimate of the impact of the programme on those who actually receive the treatment (or of ‘the effect of the treatment on those who actually received the treatment’).

The availability of repeated observations on non-eligible households in treatment areas before and after the start of the programme, also offers the opportunity to

examine the potential effects of the programme on the non-eligible households residing in the treatment communities. For example, non-eligible households in treatment localities may alter their behaviour (such as, work less or withdraw their children from school) in anticipation that such actions may qualify them for the programme. An evaluation of the extent to which the programme has had some indirect effects on the outcome indicator among non-eligible households in treatment areas can also be conducted by estimating a regression similar to Equation (1) but restricted to the sample of non-eligible households ($E = 0$).

Note that β_T is expected to be insignificantly different from zero (that is, pre-programme differences prior to programme implementation are expected to be zero) and the interaction terms represent the impact of being in a treatment community on work participation after programme implementation. The different intercept α terms capture the point that participation in work may vary (for reasons unrelated to PROGRESA) during the first round of the sample.

In estimating the impact of the programme on adult leisure time, we are limited by the fact that there is only one round of data on time allocation (June 1999). Given this constraint we estimate the model:

$$L(i) = \alpha_0 + \beta_T T(i) + \sum_{j=1}^J \beta_j X_j(i) + \varepsilon_i \quad (7)$$

where $L(i)$ measures leisure time of individual i , $T(i)$ represents a binary variable equal to 1 if individual i lives in a treatment community and 0 otherwise, and $X_j(i)$ represents the vector of j control variables for individual i (described in section 4). In this model the estimate of the coefficient β_T is the 'cross-sectional difference' (CSDIF) estimate of programme impact.

IV. The Impact of PROGRESA on Adult Labour Force Participation and Leisure

Adult Labour Force Participation

To estimate the impact on labour force participation, we use the data as our baseline round and three post-programme rounds of November 1998, June 1999 and November 1999.

The dependent variable $Y(i, t)$ in Equation (1) is specified by a binary variable indicating whether an individual i works in the labour market in period t . Specifically, a person is classified as working in the labour market ($Y(i, t) = 1$) if he/she reported having worked during the previous week (whether paid or unpaid). In addition, we take into account a follow-up question to capture individuals who may engage in informal activities, which the respondent may not have initially considered as work. This question collects information about participation in: (a) selling a product; (b) helping in family business; (c) making products to sell; (d) washing, cooking or ironing; and (e) working in agriculture activities or caring for animals. We also include as working those individuals who respond that they engage in any of these activities. It is important to keep in mind that domestic activities are not included in this definition of work.

We also consider two other outcome variables, salaried work and non-salaried work and estimate the impact of PROGRESA on each category. The distinction between salaried and non-salaried¹⁰ work is made through what a worker reports as his/her occupational position. Workers who report that they were daily agricultural workers or non-agricultural employees are considered as salaried workers. All other workers are classified separately and include self-employed workers, business owners, unpaid workers and *ejidatarios*.

Equation (1) is estimated separately for males and females. In addition, we conduct the empirical analysis separately for five age groups (18–24 years, 25–34, 35–44, 45–54 and, 55 and over) and for the group of all adults. In general, the labour force participation of women of all ages is quite low (for no age group do overall labour force participation rates exceed 18%). In addition, the majority of women who do work tend to participate in unsalaried activities: this is particularly true of women over the age of 35. It is interesting to note the decreasing relative participation in salaried work versus unsalaried activities with the age of women. Men, on the other hand, show a very high labour force participation rate, over 90 per cent for men between the ages of 24–55. The majority of men are salaried workers, although the percentage in salaried work tends to decrease with age. For male workers over the age of 55, almost half participate in non-salaried activities.

In particular, our sample includes 183,646 individual observations (89,207 observations on adult males and 94,439 observations on adult females) for four rounds of the PROGRESA evaluation sample (November 1997, October 1998, June 1999 and November 1999) which are divided by treatment and control localities. The analysis is first carried out on the sample of those who are classified as ‘poor’,¹¹ and thus eligible to receive the programme if they live in treatment localities ($E = 1$). We then re-do the same estimation on the sample of those who are classified as non-eligible ($E = 0$).

The vector X in Equation (1) consists of individual and demographic composition variables. In particular, we include as individual characteristics his/her age, age squared, marital status, whether he/she is head of the household, speaks an indigenous language, and his/her level of education. The demographic composition variables include the number of children aged 0–2 and 3–5, boys and girls between 6–7 years old, 8–12, and 13–18, men and women aged 19–54, and men and women over the age of 55. At the community level, the model also contains a variable measuring distance to the ‘*cabecera municipal*’ which is an indicator of distance to the governing centre of the municipality (and likely the largest locality of the municipality).

Given that dependent variable is binary, Equation (1) is estimated with a probit model. Table 2 presents the 2DIF estimates (summarised by the parameter β_{TR} in Equation 1) of the impact of PROGRESA on the probability of working of male and female adults for the sample of individuals from eligible households ($E = 1$).¹² The clustering of the households within villages implies that the household-specific error terms $\eta(i, t)$ are likely to be correlated within each village (as well as across time). Thus, standard errors were estimated taking into account the clustered nature of the sample. The results are presented showing the initial level of participation in work activities (that is prior to programme implementation) and, the impact estimates for each round of the survey carried out after programme implementation. The impact

Table 2. The impact of PROGRESA (2DIF estimates) on the probability of working among eligible (E = 1) adults

Age group	Pre-programme labour force	Impact on males						Impact on females												
		October 1998			June 1999			October 1998			June 1999									
		Coef.	SE	t-stat	Coef.	SE	t-stat	Coef.	SE	t-stat	Coef.	SE	t-stat							
<i>All work</i>																				
18-24	0.86	-0.005	0.018	-0.32	-0.011	0.019	-0.58	0	0.019	0.04	0.18	-0.022	0.016	-1.3	-0.037**	0.015	-2.23	-0.02	0.017	-1.13
25-34	0.94	-0.003	0.01	-0.32	-0.014	0.01	-1.45	0.001	0.009	0.2	0.16	-0.012	0.015	-0.77	-0.019	0.013	-1.33	-0.012	0.015	-0.78
35-44	0.95	0.005	0.009	0.54	0.009	0.009	0.95	0.013	0.008	1.39	0.18	-0.014	0.013	-0.99	0.011	0.018	0.61	-0.017	0.016	-0.99
45-54	0.94	0	0.013	0.01	-0.003	0.014	-0.26	0.014	0.011	1.18	0.18	-0.038**	0.014	-2.36	-0.03*	0.014	-1.91	-0.012	0.02	-0.61
55+	0.78	-0.008	0.019	-0.44	-0.02	0.023	-0.89	0	0.019	0	0.15	0.002	0.014	0.21	0.011	0.013	0.83	0.032**	0.017	1.96
All adults	0.89	-0.003	0.009	-0.36	-0.007	0.009	-0.81	0.006	0.008	0.7	0.17	-0.014	0.011	-1.24	-0.013	0.01	-1.18	-0.006	0.012	-0.5
<i>Salaried work</i>																				
18-24	0.64	0.04*	0.025	1.78	0.021	0.026	0.8	0.023	0.03	0.76	0.1	-0.003	0.012	-0.29	-0.026**	0.011	-2.08	-0.016	0.011	-1.34
25-34	0.7	0.034	0.025	1.27	0.037*	0.021	1.65	0.017	0.026	0.67	0.07	0.005	0.008	0.6	-0.003	0.008	-0.33	0.006	0.009	0.76
35-44	0.66	0.044*	0.025	1.65	0.034	0.025	1.3	0.014	0.029	0.49	0.07	0.002	0.009	0.31	0.006	0.01	0.63	-0.003	0.009	-0.37
45-54	0.62	0.054*	0.031	1.66	0.038	0.031	1.19	0.043	0.031	1.36	0.05	-0.001	0.01	-0.16	-0.005	0.009	-0.56	0.008	0.012	0.68
55+	0.43	0.04	0.034	1.18	0.025	0.031	0.79	0.019	0.033	0.58	0.04	0.005	0.007	0.77	0.006	0.008	0.85	0.013*	0.008	1.74
All adults	0.61	0.045	0.023	1.87	0.034	0.021	1.56	0.025	0.023	1.06	0.07	0.002	0.005	0.36	-0.004	0.005	-0.74	0.001	0.006	0.2
<i>Self-employed/family business</i>																				
18-24	0.21	-0.035*	0.018	-1.8	-0.017	0.02	-0.85	0	0.021	0.01	0.07	-0.013	0.009	-1.33	-0.004	0.01	-0.37	0.008	0.014	0.64
25-34	0.24	-0.03	0.022	-1.27	-0.048	0.017	-2.57	-0.01	0.023	-0.46	0.09	-0.008	0.01	-0.76	-0.01	0.008	-1.18	-0.012	0.011	-1.05
35-44	0.29	-0.036	0.024	-1.43	-0.02	0.023	-0.82	0.02	0.028	0.07	0.11	-0.013	0.008	-1.48	0.02	0.016	1.35	-0.008	0.012	-0.67
45-54	0.31	-0.047	0.03	-1.46	-0.037	0.029	-1.2	-0.023	0.029	-0.79	0.13	-0.022*	0.01	-2.22	-0.018	0.01	-1.53	-0.004	0.016	-0.29
55+	0.35	-0.031	0.027	-1.13	-0.029	0.024	-1.18	-0.002	0.027	-0.1	0.1	-0.002	0.01	-0.2	0.007	0.01	0.72	0.018	0.016	1.22
All adults	0.28	-0.035	0.019	-1.75	-0.031	0.017	-1.78	-0.007	0.02	-0.38	0.1	-0.011	0.007	-1.46	-0.002	0.008	-0.27	0	0.01	-0.07

Notes: ***significant at 1 per cent level; **significant at 5 per cent level; *significant at 10 per cent level. See text for more details.

from each round should be interpreted as the percentage point difference from the pre-programme level (not from the previous round). In other words, the estimates reported represent the marginal effects of being in a household eligible for PROGRESA benefits on the probability of being in the labour force.¹³

Beginning with men, the results of the impact of PROGRESA on overall participation levels show little impact. Moreover, irrespective of the age group examined, participation in PROGRESA seems to have no impact on labour market participation.

Looking at the decomposition between salaried work and other types of works, there are some impacts, particularly in the November 1998 round – the first round after PROGRESA was implemented in these communities. In this round, there is a universal increase, for all age groups, in the probability of working in salaried work and a corresponding decrease in the probability of working in non-salaried work. These effects remain present in the next round of the data (June 1999) only for men aged 25–34 and disappear by the fourth round of the survey. The results seem to suggest that, at least initially, families may have used some part of the grants to seek work in salaried activities and to reduce their participation, perhaps, in less profitable family enterprises. This impact, however, appears to disappear over time.

For women, the results show few overall impacts of PROGRESA on participation in the labour market. For women in the age group 45–54, there is a significant reduction in participation according to the first after programme round of the ENCEL, although this impact does not hold up over time. As with men, there is also a significant reduction in the probability of participating in non-salaried work activities in the first after programme round but, again, these effects do not hold up over time. In short, these data do not show particularly significant or lasting effects of PROGRESA on labour market participation. Rather, they are consistent with a story that PROGRESA does not affect participation of men and women.

In table 3, we present the results for the sample of individuals who are classified as non-eligible for the programme (or $E = 0$). Even though these individuals are not in PROGRESA (which only covers the ‘poor’ individuals in treatment localities), the programme may have been generating *spillover* effects; for example, households in PROGRESA localities who are not poor could have revised their labour force participation choices in order to become eligible for the programme. However, our results suggest that PROGRESA does not have any significant effect on participation on individuals who are not poor; the only coefficients that are significant (only at 10% level) are all negative, but very small in magnitude.

Leisure

Our measure of leisure time is constructed as a residual, that is as the difference between 24 hours and the time spent on all reported activities. In particular, we use the time use module (present only in the June 1999 round) with information on 18 activities¹⁴ carried out during the previous day; the previous day as reference period is not particularly ideal, as for some individuals, the survey may refer to a day which was not ‘typical’ of normal activities. Additionally, many activities may be activities which are done infrequently (that is, not daily). The survey was carried out this way as it was thought that a ‘previous day’ reference period would reduce recall bias,

Table 3. The impact of PROGRESA (2DIF estimates) on the probability of working among non-eligible (E=0) adults

Age group	Pre-programme labour force participation	Impact on males						Pre-programme labour force participation	Impact on females											
		October 1998			June 1999				October 1998			June 1999								
		Coef.	SE	t-stat	Coef.	SE	t-stat		Coef.	SE	t-stat	Coef.	SE	t-stat						
		<i>All Work</i>																		
18-24	0.88	0.003	0.026	0.15	-0.029	0.029	-1.03	-0.016	0.029	-0.57	0.29	0.001	0.03	0.05	0.03	-0.22	-0.003	0.034	-0.1	
25-34	0.95	-0.021	0.021	-1.03	-0.036*	0.023	-1.74	0.002	0.02	0.11	0.32	0.015	0.033	0.47	0.039	0.037	1.08	-0.01	0.034	-0.3
35-44	0.96	-0.022	0.023	-1.1	-0.008	0.02	-0.43	0.004	0.02	0.21	0.26	-0.018	0.031	-0.57	0.012	0.031	0.39	-0.061**	0.028	-1.96
45-54	0.96	-0.032	0.026	-1.34	-0.016	0.024	-0.85	0.015	0.019	0.71	0.21	0.018	0.028	0.69	0.007	0.028	0.25	-0.032	0.021	-1.42
55+	0.88	-0.06	0.024	-2.26	-0.026	0.028	-0.97	-0.017	0.027	-0.65	0.18	-0.034*	0.016	-1.9	-0.031*	0.017	-1.67	-0.035*	0.019	-1.62
All adults	0.92	-0.014	0.011	-1.23	-0.024	0.012	-2.03	-0.007	0.011	-0.63	0.27	-0.01	0.017	-0.59	-0.002	0.017	-0.14	-0.03	0.017	-1.71
<i>Salariat work</i>																				
18-24	0.57	0.014	0.037	0.39	-0.015	0.047	-0.32	-0.029	0.04	-0.73	0.17	0.003	0.024	0.16	0.021	0.029	0.74	0.024	0.029	0.87
25-34	0.58	0.028	0.034	0.83	0.023	0.038	0.61	0.027	0.04	0.68	0.15	0.029	0.025	1.23	0.015	0.027	0.57	0.007	0.024	0.31
35-44	0.54	0.042	0.04	1.02	0.046	0.04	1.12	0.05	0.048	1.01	0.11	0.01	0.02	0.5	-0.01	0.019	-0.5	-0.007	0.019	-0.35
45-54	0.47	-0.035	0.051	-0.7	-0.032	0.0511	-0.63	0.028	0.511	0.55	0.08	0.001	0.015	0.08	0.004	0.016	0.25	-0.02*	0.009	-1.79
55+	0.37	0.013	0.044	0.3	-0.01	0.041	-0.24	-0.032	0.041	-0.77	0.04	0.001	0.009	0.14	-0.003	0.009	-0.55	-0.003	0.008	-0.43
All adults	0.5	0.016	0.028	0.59	0.002	0.03	0.1	0.001	0.029	0.04	0.11	0.005	0.009	0.57	0.002	0.011	0.24	-0.002	0.009	-0.3
<i>Self-employed/family business</i>																				
18-24	0.31	-0.02	0.03	-0.64	-0.01	0.033	-0.3	0.009	0.035	0.29	0.11	0.013	0.021	-0.7	-0.016	0.014	-1.01	-0.019	0.016	-1.08
25-34	0.38	-0.043	0.029	-1.41	-0.052	0.031	-1.57	-0.019	0.036	-0.52	0.16	0	0.021	-0.04	0.025	0.027	1.01	-0.015	0.02	-0.71
35-44	0.42	-0.054	0.037	-1.42	-0.046	0.037	-1.2	-0.037	0.045	-0.8	0.18	-0.015	0.023	-0.62	0.021	0.025	0.81	-0.044*	0.02	-1.85
45-54	0.49	0.01	0.046	0.22	0.02	0.047	0.43	-0.009	0.048	-0.18	0.17	0.011	0.023	0.49	0.006	0.021	0.3	-0.002	0.02	-0.14
55+	0.51	-0.017	0.038	-0.45	-0.01	0.038	-0.27	0.021	0.039	0.54	0.18	-0.026*	0.012	-1.89	-0.02	0.014	-1.28	-0.026	0.015	-1.47
All adults	0.42	-0.026	0.024	-1.06	-0.02	0.025	-0.79	-0.002	0.027	-0.1	0.16	-0.005	0.012	-0.45	-0.001	0.012	-0.14	-0.022	0.011	-1.85

Notes: ***significant at 1 per cent level, **significant at 5 per cent level, *significant at 10 per cent level. See text for more details.

given the large number of activities included in the questionnaire. The control variables included are the same as those in the labour force participation estimation (discussed above).

Table 4 presents the CSDIF estimates of the effect of PROGRESA on adult leisure (summarised by parameter β_T in Equation (7)). Note that one hypothesis of the impact of PROGRESA on leisure is that, if the PROGRESA transfers are perceived as strict income transfers, and leisure is a normal good, then one might expect leisure to increase with PROGRESA. Nevertheless, the structure of the grants which reduce the price of schooling of children and thus may reduce the work of children, may imply that overall hours dedicated to household production work (previously done by children) might increase. This would then imply that the programme would have an ambiguous effect on the leisure time of adults, especially women. For example, it is conceivable that complying with the programme's requirements might decrease the leisure time of women, as they attempt to substitute for the time children used to allocate to home production activities.¹⁵

Overall, the results do not show significant impacts of PROGRESA on the leisure time of men or women. There are some small negative impacts of PROGRESA on leisure for men for one age group, namely men aged 18–24, which corresponds to increases in work for this group of men of about 0.3 hours daily, or about 2 hours weekly. Nevertheless, there are no significant impacts on any other age groups for males. The results for women are insignificant in all specifications and for all groups. Accordingly, we can say that there is not much evidence to support the hypothesis that PROGRESA has reduced the leisure time of men and women. There is certainly no evidence to support that leisure time has increased under PROGRESA.¹⁶

V. The Impact of PROGRESA on Poverty

The results presented so far on the impact of the programme on adult labour market participation and leisure suggest that the programme has no adverse effects on labour income. Thus, *ceteris paribus*, the cash transfers received by programme beneficiaries are likely to increase household income at least among beneficiary families. However, the extent to which this occurs depends on the size of the direct and indirect costs associated with participation in the programme. In addition, the income of non-eligible households may be adversely affected to the extent that

Table 4. The impact of PROGRESA on leisure time of programme eligible (E = 1) adults

Age group	Pre-programme daily hours	Men impact June 1999			Pre-programme daily hours	Women impact June 1999		
		Coef.	SE	t-stat		Coef.	SE	t-stat
18–24	16.24	–0.321*	0.169	–1.9	17.18	0.026	0.087	0.3
25–34	14.69	0.122	0.122	1	16.17	–0.236	0.148	–1.6
35–44	14.64	–0.061	0.087	–0.7	16.65	–0.016	0.160	–0.1
45–54	14.72	0.06	0.200	0.3	17.44	0.023	–0.230	–0.1
55+	16.63	–0.144	0.206	–0.7	19.21	0.09	0.150	0.6

Note: *significant at 10 per cent level.

non-eligible households believe that a lower household income increases their chances of becoming eligible for the programme.¹⁷ Estimating the impact of the programme on the poverty rate provides a useful summary measure of how the programme affects the income of both eligible and non-eligible households in the treatment localities relative to those in the control localities.

The measure of welfare used in our analysis is income per capita.^{18,19} For households in treatment villages receiving PROGRESA cash transfers, total income per month was adjusted upwards by the cash transfer per month received by the household.²⁰ The actual amount of cash transfers received per month was obtained from the records of payments sent out each month since May 1998 by the PROGRESA administration headquarters in Mexico City. The monthly income measure calculated for each round of the survey was then converted to November 1998 pesos by dividing by the corresponding adjustment ratio of the national consumer price index. We calculate different poverty measures using two different poverty lines: the value of the basic food basket (*canasta basica*) and the median of the per capita value of consumption in November 1998. The first poverty line (basic food basket) yields a baseline poverty rate of 82.16 per cent in the treatment localities (and 81.99% in the control localities). The median of per capita consumption in November 1998 yields a baseline headcount poverty rate $P(0)$ of 55.44 per cent in the treatment localities (53.16% in the control localities), which is slightly below the fraction of the population in the treatment villages that actually received the benefits of the programme (see section II).

Poverty is measured along the lines suggested by Foster et al. (1984) (henceforth FGT). The FGT poverty measures are summarised by the formula:

$$P(\alpha) = \left(1/N\right) \sum_{i=1}^q \left(\frac{z - y_i}{z}\right)^{\alpha}, \quad (8)$$

where N is the total number of households, y_i is the per capita income of the i th household, z is the poverty line, q is the number of poor individuals, and α is the weight attached to the severity of household poverty (or the distance from the poverty line). When $\alpha=0$, the FGT measure collapses to the Headcount Index, or the percentage of the population that is below the poverty line. When $\alpha=1$ the FGT measure gives the *poverty gap* $P(1)$, a measure of the average depth of poverty. When $\alpha=2$, the FGT index becomes the *severity of poverty* index. The $P(2)$ measure assigns more weight to individuals that are further away from the poverty line.²¹

As described in section III, relying on reported household income allows one to obtain the difference-in-differences (2DIF) estimate of the impact of the programme on poverty, which compares the change in a poverty measure in treatment villages to the changes in the corresponding poverty measure in control villages. In addition to controlling for macroeconomic shocks common to both treatment and control localities, this estimate allows one to account for any pre-existing differences in poverty between control and treatment localities and thus yield 'cleaner' estimate of the impact of the programme on poverty.

The regression equation behind the estimation of PROGRESA's impact on poverty is:

$$P(i, t, \alpha) = \beta_0 + \beta_T T(i) + \beta_{R2}(R2) + \beta_{TR2}(T(i) \times R2) + \beta_{R3}(R3) + \beta_{TR3}(T(i) \times R3) + \beta_{R4}(R4) + \beta_{TR4}(T(i) \times R4) + \eta(i, t) \quad (9)$$

where the left-hand side variable $P(i, t, \alpha)$ is defined as,

$$P(i, t, \alpha) = \left(\frac{z - y(i, t)}{z} \right)^\alpha \times Poor(i, t), \quad (10)$$

where $y(i, t)$ denotes the income of household i in period/round t , z is the poverty line used, α takes on the values 0, 1, and 2, and $Poor(i, t)$ is a binary variable taking the value of 1 if $y(i, t) \leq z$, and equal to 0 otherwise. Based on the specification of the regression Equation (9), the intercept term β_0 is the estimate of the poverty rate (headcount ratio, poverty gap, or the severity of poverty) in the control localities in the baseline round, while $\beta_0 + \beta_T$ is the corresponding estimate of poverty in the treatment localities (in the baseline round).²² As discussed above, the estimates of the parameters β_{TR2} , β_{TR3} , and β_{TR4} , are the 2DIF estimates of the impact of the programme in rounds 2, 3 and 4 of the survey.

In Table 5 we present the estimates of the parameters of Equation (9) along with standard errors adjusted for the clustering of households within villages.²³ The negative and strongly significant estimates of β_{TR2} , β_{TR3} , and β_{TR4} imply that PROGRESA had a significant impact in reducing poverty between November 1997 and November 1999. For example, using the 50th percentile of the value of consumption per capita as a poverty line, suggests that the headcount poverty rate declined by around 4.88 per cent between November 1997 and November 1998, and by 18.11 per cent in the November 1999 in treatment areas (using as base the 55.44% headcount poverty rate in treatment localities in November 1997). During the same period, and using as base the corresponding value of the poverty gap and squared poverty gap indices in treatment areas in November 1997, the poverty gap measure declined by 37.40 per cent, and the severity of poverty measure (squared poverty gap) declined by 47.42 per cent. The higher impacts of the programme in reducing poverty over time are consistent with the findings of Gertler et al. (2006), who demonstrated that rural households increased their investments in micro enterprises and agricultural activities which, in turn, improved the households ability to generate income.

In general, these estimates are remarkably in line with the estimates obtained using *ex-ante* simulations. These simulations are based on the predicted consumption of each household in the evaluation sample in November 1997, adding the maximum amount of PROGRESA cash transfers an eligible household could receive assuming full compliance with the programme's requirements (see Skoufias et al., 2001). In particular, the results obtained from the simulated impact of PROGRESA's cash transfers show that the headcount ratio is reduced by about 10 per cent through the supports of PROGRESA. The poverty gap and severity of poverty measures, which

Table 5. The impact of PROGRESA on poverty using two different poverty lines

Poverty line: Canasta basica in November 1997					Poverty line: Median of November 1998 (Consumption p.c.)						
Variable	Coef.	SE	t-value	p-value	Impact*	Variable	Coef.	SE	t-value	p-value	Impact*
(a) Headcount ratio P(0)											
T	0.0017	0.012	0.13	0.893		T	0.0228	0.019	1.23	0.219	
R2	0.0183	0.007	2.49	0.013		R2	0.0379	0.012	3.28	0.001	
R2 × T	0.0007	0.009	0.08	0.938	0.09	R2 × T	-0.0271	0.014	-1.89	0.059	-4.88
R3	0.0429	0.009	4.91	0		R3	0.1120	0.011	10.46	0	
R3 × T	-0.0060	0.011	-0.57	0.568	-0.73	R3 × T	-0.0545	0.014	-4.03	0	-9.83
R4	0.0392	0.006	6.29	0		R4	0.0533	0.012	4.39	0	
R4 × T	-0.0207	0.009	-2.26	0.024	-2.52	R4 × T	-0.1004	0.016	-6.3	0	-18.11
_cons	0.8199	0.010	78.2	0		_cons	0.5316	0.015	36.44	0	
(b) Poverty gap P(1)											
T	0.0215	0.014	1.53	0.126		T	0.0333	0.014	2.32	0.021	
R2	0.0399	0.008	4.7	0		R2	0.0425	0.010	4.36	0	
R2 × T	-0.0284	0.011	-2.61	0.009	-5.70	R2 × T	-0.0457	0.013	-3.6	0	-15.92
R3	0.0992	0.008	11.97	0		R3	0.1248	0.009	14.19	0	
R3 × T	-0.0445	0.011	-4.2	0	-8.95	R3 × T	-0.0717	0.012	-6.08	0	-24.99
R4	0.0375	0.008	4.59	0		R4	0.0306	0.009	3.33	0.001	
R4 × T	-0.0794	0.012	-6.74	0	-15.94	R4 × T	-0.1074	0.014	-7.71	0	-37.40
_cons	0.4763	0.011	45	0		_cons	0.2538	0.010	26.6	0	
(c) Severity of poverty P(2)											
T	0.0287	0.014	2.1	0.037		T	0.0346	0.013	2.65	0.008	
R2	0.0412	0.009	4.58	0		R2	0.0382	0.009	4.18	0	
R2 × T	-0.0393	0.012	-3.37	0.001	-10.96	R2 × T	-0.0488	0.012	-4	0	-24.21
R3	0.1151	0.008	13.93	0		R3	0.1257	0.008	15.03	0	
R3 × T	-0.0616	0.011	-5.61	0	-17.15	R3 × T	-0.0774	0.011	-6.73	0	-38.42
R4	0.0325	0.008	3.83	0		R4	0.0223	0.008	2.7	0.007	
R4 × T	-0.0938	0.013	-7.36	0	-26.13	R4 × T	-0.0955	0.014	-7.03	0	-47.42
_cons	0.3301	0.010	34.68	0		_cons	0.1669	0.008	21.11	0	

Note: *The columns labeled 'impact' contain the 2DIF estimate of the programme's impact in each round as a per cent of the respective poverty measure in the treatment localities in the baseline round.

place greater weight on the poorest households within the population in poverty, show that the level of poverty according to the poverty gap is reduced by 30 per cent whereas the severity of the poverty index (squared poverty gap) is reduced by 45 per cent.

The choice of a poverty line is a major concern when poverty measures are estimated. In addition to estimating programme impact on poverty based on two alternative poverty lines, we also conduct tests of stochastic dominance, up to order three, between the distribution of monthly income in treatment and control areas for each round. As far as the programme impact on poverty is concerned, stochastic

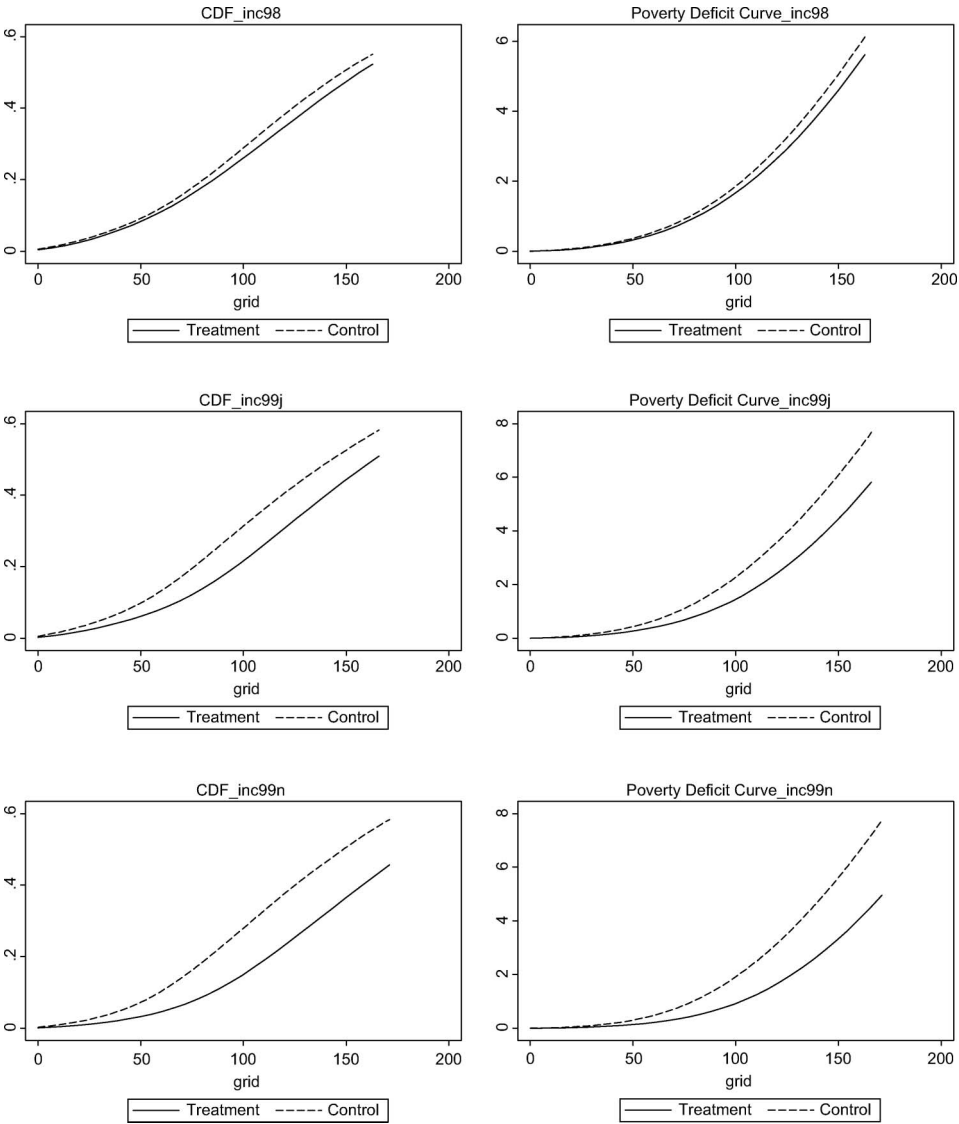


Figure 2. Estimated distribution function and poverty deficit curve. Note: x-variable is income

dominance of one distribution on the other can have important implications. The order of stochastic dominance achieved leads to different conclusions as regards to poverty measures considered, with first-order stochastic dominance being the most stringent criteria. For instance, if the distribution of income in treatment areas first-order stochastically dominates that in control areas, this implies that all the poverty measures we are considering [headcount ratio $P(0)$, poverty gap $P(1)$, and squared poverty gap $P(2)$] will always show less poverty in treatment areas no matter which poverty line is chosen. Higher orders of stochastic dominance mean that poverty will be less in treatment areas regardless of the poverty line only according to a smaller set of poverty measures; for example, second-order stochastic dominance implies only $P(1)$ and $P(2)$ showing less poverty in treatment areas regardless of the poverty line (for more details see Duclos and Araar, 2006). In brief, stochastic dominance is explored here in order to understand whether there is evidence of a programme impact on poverty that is a robust choice of a poverty line.

Results from our test of stochastic dominance (for details see Davidson and Duclos, 2000; test results are in Appendix 1) add some robustness to the evidence above. In particular, we find that the distribution of income for population of treated households first-order stochastically dominates that of control group in the June and November 1999 rounds (while for round November 1998 we find third-order stochastic dominance; see also Figure 2 where we report the estimated CDF's and poverty deficit curves²⁴). An implication of these results is that in June and November 1999 $P(0)$ (and so $P(1)$ and $P(2)$) will be always smaller in treatment than in the control group regardless of the poverty line chosen.

In conclusion, since we find the strongest evidence of poverty reduction when we analyse poverty gap and severity of the poverty index (poverty gap squared), which put greater weight on the poorest of the poor, both the simulation and the actual ex-post results suggest that the largest reductions in poverty of PROGRESA are being achieved in the poorest of the poor population.

VI. Conclusions

Conditional cash transfer (CCT) programmes aim to alleviate current poverty through monetary and in-kind benefits, as well as reduce future levels of poverty by encouraging investments in education, health and nutrition. The success of CCT programmes at reducing current poverty depends on whether, and the extent to which, cash transfers affect adult work incentives.

Based on the experimental design of PROGRESA's evaluation sample, our findings yield a very consistent answer. PROGRESA does not have any significant effect on adult labour supply choices. In particular, the results show that there has been no particular reduction in labour market participation rates, as may have been predicted by some economic models of behaviour. There is some evidence that individuals, at least right after they started to receive the transfers, may have used part of the grants to seek work in salaried activities and to reduce their participation in perhaps less profitable family enterprises. This impact, however, appears to disappear over time (see the results for the final round, November 1999). In addition, there is not much evidence to support the hypothesis that PROGRESA beneficiaries use their transfers to 'buy' more leisure.

The success of CCT programmes at reducing current poverty depends on whether, and the extent to which, cash transfers affect adult work incentives. In particular, policy makers have concerns about possible adverse effects on labour supply, such as a disincentive to work for eligible households or even for non-eligible households (that might hope to become eligible for the programme in future rounds of expansion of the programme). It is then a welcome finding that PROGRESA is not having a significant effect on adult labour supply (and, thus, any adverse effect). We also stress that the programme does not have the objective of affecting adult labour supply, with this suggesting that its design is working properly.

Our findings on adult work incentives are reinforced further by the result that PROGRESA leads to a substantial reduction in current poverty. The poverty reduction effects are stronger for the poverty gap and severity of poverty measures, which put greater weight on the poorest of the poor, and our evidence suggests that these estimated poverty measures are robust to the choice of different poverty lines. As an additional piece of evidence, we notice that our results on the PROGRESA impact on poverty rates are remarkably in line with the *ex-ante* simulated impact of PROGRESA's cash transfers (as in Skoufias et al., 2001). Thus, *ex-ante* simulations that ignore or assume away behavioural responses to the transfer (as in Bourguignon et al., 2003) are likely to provide good estimates of the *ex-post* impact on poverty.

Acknowledgements

The authors are grateful to two anonymous referees for constructive comments, Claudia Aburto for excellent research assistance and to Alessandro Tarozzi for useful suggestions. The findings, interpretations and conclusions in this paper are entirely those of the authors and they do not necessarily reflect the view of the World Bank.

Notes

1. In most CCT programmes, the eligibility status of beneficiary households is, in theory, re-examined every few years. For example, in the PROGRESA programme of Mexico, the eligibility status of households was supposed to be reviewed within three years after a household's entry into the programme. In fact, more than five years elapsed before any effort was made to revise the list of beneficiaries.
2. Skoufias (2005) provides a detailed discussion of PROGRESA, the evaluation design and a summary of the impacts of the programme estimated by a large team of researchers.
3. The average monthly transfers during the 12 month period from November 1998 to October 1999 are around 197 pesos per beneficiary household per month (expressed in November 1998 pesos). The calculation of this average includes households that did not receive any benefits due to non-adherence to the conditions of the programme, or delays in the verification of the requirements of the programme or in the delivery of the monetary benefits. On average, households receive 99 pesos for food support, and 91 pesos for the educational grant. For more details, see Hoddinott and Skoufias (2004).
4. Before the establishment of PROGRESA, previous government interventions in the areas of education, health and nutrition in the rural sector of the country consisted of many programmes each intervening separately in health, education or nutrition with little prior coordination or consideration of the potential synergies that could result from a better coordinated and simultaneous intervention.

5. Behrman and Todd (1999) conducted a careful investigation of the extent to which the selection of localities into treatment and control groups can be considered as random. Their analysis did not reveal any significant differences between village means for more than 300 variables in treatment and controls.
6. For more details see Skoufias (2005).
7. The March 1998 baseline survey was not used because it did not include household income and labour supply information for adults.
8. Note that control households started receiving cash benefits in December 1999. Households are first incorporated into PROGRESA, meaning that they are given all the necessary forms and informed of all the programme requirements. A few months later, the cash benefits are sent out by the PROGRESA administration headquarters.
9. Behrman and Todd (1999) state in their investigation of whether assignment to treatment and control groups can be considered random that formal tests of equality between the distributions of various characteristics generally do not reject the hypothesis when the test is performed on locality means. However, when the test is performed on household level data, they find many more rejections of the null than would be expected by chance given standard significance level. This motivates the choice of including controls in our regressions and employ a 2DIF approach.
10. In preliminary analysis, we considered separating non-salaried workers between self-employed workers and unpaid family workers. Nevertheless, the proportions of individuals participating in each of these activities are quite small for all age groups, and the distinction between these activities is often blurred so that we prefer to aggregate these groups in the impact analysis.
11. This is the poor status after the *densificación*, the revision of the eligibility that raised the number of households eligible for the programme from 52 to 78 per cent. It has to be noticed that the fraction of households that actually ended up receiving the PROGRESA cash transfers during the two-year interval covered by the evaluation sample is just under 65 per cent, due to administrative errors and delays in the final registration of beneficiary households.
12. The complete set of parameter estimates is available directly from the authors upon request.
13. The estimates reported were obtained using the 'dprobit' command in STATA v7.0. They can be easily converted into percentage changes or elasticities by dividing the marginal effect by the pre-programme level, both reported in Table 2.
14. Activities are: Working for salary or wage, in own business or family land; Attending school; Doing homework after school; Community work; Voluntary work for neighbors or other relatives; Purchasing food or other products for the household (HH); Sewing, making clothes for HH members; Taking HH members to school, clinic, or work; Cleaning house; Washing and ironing clothes for HH members; Preparing food; Fetching water, firewood or throwing out trash; Taking care of animals; Taking care of small children, elderly and sick; Making HH repairs; Transportation time to work, school, market and so forth; Other activities.
15. Skoufias and Parker (2001) and Schultz (2004) study the impact of PROGRESA on the work time and school attendance of school-age children. In particular, Skoufias and Parker (2001) find significant increases in the school attendance of boys and girls that are accompanied by significant reductions in the participation of boys and girls in work activities.
16. Note, however, that the results on leisure do not necessarily suggest that there has been no reallocation of time between work activities for adults. For instance, there may have been a substitution towards more time in domestic work and less time in market work. (for results on this issue see Parker and Skoufias, 2000).
17. Our measure of income is based on reported income, with this bringing into the picture all the possible biases due to misreporting (especially underreporting) of income. As stated above, eligibility for the programme was decided based on socioeconomic data collected in the 1997 baseline household census (ENCASEH) before the start of the programme. From our knowledge, households were not told that the ENCASEH information was going to be used to discriminate eligibility. In addition to this, households did not have to apply for the programme but this was universally given to all households identified as eligible in the treatment localities. This suggests that there might not be a clear and strong incentive to misreport income. In general, evidence on this issue is scarce due to the severity of data requirements. Martinelli and Paker (2006) study this issue with data from the urban evaluation sample of *Oportunidades*; it has to be noticed that the rural and urban component of ProgresA have quite different characteristics: for example, households need to apply for the programme in the urban

version and find that while underreporting is widespread, over reporting is also common in goods that may have a 'status' value.

18. The absence of reliable information on household consumption prior to the start of the programme precluded the use of consumption as a measure of the poverty impacts of the programme. For more details on the consumption impacts of PROGRESA see Hoddinott and Skoufias (2004).
19. We do not consider value of leisure in our definition of income (that is, our income is not full income). Our definition of income is consistent with that used for deciding eligibility for the programme. It is important to stress that the effect of the programme on poverty may be depending upon the definition of income; for example, PROGRESA would have *a priori* a positive effect on poverty in case we employed a full income approach (and so we considered the value of leisure).
20. Many studies have considered whether the introduction of public transfers affects private transfers among the households targeted by the public scheme. For example, Cox and Jimenez (1990) argue that public cash transfers may reduce the amount of private transfers to low-income households so that the net-income effect may be significantly less than the value of the public transfer. Albarran and Attanasio (2003) study this issue with PROGRESA data and find that the programme does crowd out private transfers: both the likelihood of receiving a transfer and the amount received conditional on receiving private transfers are significantly and negatively affected by the programme.
21. FGT poverty measures are related to stochastic dominance. In particular, first-order stochastic dominance (SD) implies that all $P(\alpha)$ for $\alpha > 0$ are robust to the choice of the poverty line; the same applies for all $\alpha > 1$ for second-order SD and for all $\alpha > 2$ for third-order SD.
22. Along similar lines, $\beta_0 + \beta_{R2}$ is the poverty rate in control localities in round 2 and $\beta_0 + \beta_T + \beta_{R2} + \beta_{TR2}$ is the poverty rate in treatment localities in the same round.
23. We have also estimated the impact of the programme by symmetrically trimming the top and bottom five per cent of the sample of observations in each round so as to eliminate extreme outliers from the sample. Using the trimmed sample resulted in slightly lower impacts of the programme on poverty. Overall, however, the estimates of the impact of the programme on poverty did not change the results presented above in any substantial manner, which implies that the role of outlier observations in income is trivial.
24. The estimated CDF gives the $P(0)$ for any level of income. Poverty deficit curve is defined as the area under the CDF up to some poverty line. If the poverty deficit curve of one distribution lies above the poverty deficit curve of another, the first distribution will always have more poverty according to the poverty-gap measure, $P(1)$.

References

- Albarran, P. and Attanasio, O. (2003) Limited commitment and crowding out of private transfers: evidence of a randomized experiment. *Economic Journal*, 113(486), pp. C77–C85.
- Behrman, J. and Todd, P.E. (1999) Randomness in the experimental samples of PROGRESA (education, health, and nutrition program). Report submitted to PROGRESA, February, International Food Policy Research Institute, Washington DC.
- Blundell, R. and MaCurdy, T. (1999) Labour supply: a review of alternative approaches, in: O. Ashenfelter and D. Card (eds.) *Handbook of Labour Economics*, Vol. 3A (Amsterdam: North Holland), chap. 27, pp. 1559–1695.
- Bourguignon, F., Ferreira, F.H.G. and Leite, P.G. (2003) Conditional cash transfers, schooling, and child labor: microsimulating Brazil's Bolsa Escola program. *World Bank Economic Review*, 17(2), December, pp. 229–254.
- Cox, D. and Jimenez, E. (1990) Achieving social objective through private transfers: a review. *World Bank Research Observer*, 5(2), pp. 205–218.
- Davidson, R. and Duclos, J.-Y. (2000) Statistical inference for stochastic dominance and for the measurement of poverty and inequality. *Econometrica*, 68(6), pp. 1435–1464.
- Foster, J., Greer, J. and Thorbecke, E. (1984) A class of decomposable poverty measures. *Econometrica*, 52, 761–765.
- Gertler, P., Martinez, S. and Rubio, M. (2006) Investing cash transfers to raise long term living standards. World Bank Policy Research working paper no. 3994, World Bank, Washington DC.

- Heckman, J.J., La Londe, R. and Smith, J. (1999) The economics and econometrics of active labor market programs, in: O. Ashenfelter and D. Card (eds) *Handbook of Labour Economics*, Vol. 3A (Amsterdam: North Holland), chap. 31, pp. 1865–2097.
- Hoddinott, J. and Skoufias, E. (2004) The impact of PROGRESA on food consumption. *Economic Development and Cultural Change*, 53(1), October, pp. 37–61.
- León, M., Vos, R. and Brborich, V. (2001) ¿Son efectivos los programas de transferencias monetarias para reducir la pobreza? Evaluación de impacto del Bono Solidario en el Ecuador. Draft.
- Martinelli, C. and Parker, S.W. (2006) Deception and misreporting in a social program. Working paper 0602, Centro de Investigacion Economica, ITAM.
- Moffitt, R. (1992) Incentive effects of the US welfare system: a review. *Journal of Economic Literature*, 30(1), March, pp. 1–61.
- Parker, S. and Skoufias, E. (2000) The impact of PROGRESA on work, leisure and time allocation. Report submitted to PROGRESA, International Food Policy Research Institute, Washington DC.
- Sahn, D. and Alderman, H. (1996) The effect of food subsidies on labor supply in Sri Lanka. *Economic Development and Cultural Change*, 45, October, pp. 125–145.
- Schultz, T.P. (2004) School subsidies for the poor: evaluating the Mexican Progresa poverty program. *Journal of Development Economics*, 74(1), June, pp. 199–250.
- Skoufias, E. (2005) PROGRESA and its impacts on the welfare of rural households in Mexico. IFPRI research report no. 139, International Food Policy Research Institute, Washington DC.
- Skoufias, E. and Parker, S.W. (2001) Conditional cash transfers and their impact on child work and schooling: evidence from the PROGRESA program in Mexico. *Economia*, 2(1), Fall, pp. 45–96.
- Skoufias, E., Davis, B. and de la Vega, S. (2001) Targeting the poor in Mexico: an evaluation of the PROGRESA selection mechanism. *World Development*, 29(10), October, pp. 1769–1784.
- Stafford, F. (1985) Income-maintenance policy and work effort: learning from experiments and labor market studies, in: J.A. Hausman and D.A. Wise (eds) *Social Experimentation* (Chicago: University of Chicago Press), pp. 95–143.
- Widerquist K. (2005) A failure to communicate: what (if anything) can we learn from the negative income tax experiments? *Journal of Socio-Economics*, 34(1), pp. 49–81.

Appendix 1. Test of Stochastic Dominance and Poverty

Test Results

Z = grid point for income.

D1 = statistics for distribution if households in treatment areas and poor.

D2 = statistics for distribution if households in non treatment areas and poor.

November 1999

Minimum test point is 22.631578.

Maximum test point is 143.33333.

Order 1			
Z	D1	D2	t-statistic
22.63	0.08	0.12	3.21
28.98	0.10	0.13	2.40
35.34	0.11	0.13	1.56
41.69	0.13	0.14	0.94
48.04	0.14	0.15	0.82
54.40	0.16	0.17	0.82
60.75	0.17	0.19	1.14

67.10	0.19	0.21	1.29
73.45	0.21	0.23	0.87
79.81	0.23	0.25	0.71
86.16	0.26	0.28	1.30
92.51	0.28	0.30	1.03
98.86	0.30	0.32	0.55
105.22	0.34	0.37	1.78
111.57	0.36	0.39	1.33
117.92	0.38	0.41	1.05
124.28	0.41	0.44	1.46
130.63	0.44	0.47	1.70
136.98	0.46	0.50	1.72
143.33	0.49	0.52	1.48
Order 2			
Z	D1	D2	t-statistic
22.63	1.15	2.30	5.05
28.98	1.71	3.08	4.52
35.34	2.37	3.91	4.01
41.69	3.15	4.79	3.50
48.04	4.01	5.73	3.10
54.40	4.96	6.75	2.80
60.75	6.00	7.88	2.58
67.10	7.13	9.13	2.43
73.45	8.41	10.52	2.29
79.81	9.83	12.04	2.16
86.16	11.40	13.73	2.08
92.51	13.13	15.60	2.01
98.86	15.00	17.57	1.93
105.22	17.05	19.84	1.92
111.57	19.26	22.25	1.91
117.92	21.62	24.78	1.88
124.28	24.15	27.51	1.86
130.63	26.85	30.47	1.88
136.98	29.71	33.58	1.88
143.33	32.72	36.82	1.88
Order 3			
Z	D1	D2	t-statistic
22.63	10.89	24.48	5.52
28.98	19.91	41.56	5.23
35.34	32.82	63.76	4.91
41.69	50.33	91.36	4.57
48.04	73.05	124.71	4.24
54.40	101.51	164.29	3.94
60.75	136.29	210.72	3.68
67.10	177.93	264.70	3.45
73.45	227.21	327.05	3.26
79.81	285.06	398.63	3.10
86.16	352.41	480.38	2.95
92.51	430.24	573.48	2.83
98.86	519.53	678.79	2.72
105.22	621.25	797.41	2.62
111.57	736.53	931.05	2.55
117.92	866.31	1080.39	2.48
124.28	1011.61	1246.36	2.42

130.63	1173.52	1430.43	2.37
136.98	1353.06	1633.74	2.32
143.33	1551.26	1857.31	2.29

Dominance achieved at order 3.

June 1999

Minimum test point is 20.678703.

Maximum test point is 130.96512.

Order 1 Z	D1	D2	t-statistic
20.68	0.11	0.20	8.30
26.48	0.13	0.21	6.52
32.29	0.16	0.21	4.55
38.09	0.17	0.22	3.72
43.90	0.19	0.23	2.61
49.70	0.21	0.24	2.41
55.51	0.22	0.25	2.27
61.31	0.24	0.27	2.52
67.12	0.25	0.29	2.51
72.92	0.26	0.31	2.91
78.72	0.28	0.34	3.64
84.53	0.30	0.36	3.49
90.33	0.33	0.41	4.42
96.14	0.35	0.42	3.91
101.94	0.36	0.43	3.43
107.75	0.39	0.46	3.97
113.55	0.42	0.49	3.74
119.36	0.44	0.51	3.67
125.16	0.46	0.52	3.00
130.97	0.48	0.54	2.97

Dominance achieved at order 1.

November 1999

Minimum test point is 25.789475.

Maximum test point is 163.33334.

Order 1 Z	D1	D2	t-statistic
25.79	0.04	0.10	6.72
33.03	0.05	0.11	6.16
40.27	0.06	0.11	5.82
47.51	0.07	0.12	5.66
54.75	0.08	0.14	6.83
61.99	0.09	0.17	7.27
69.22	0.10	0.19	7.16
76.46	0.12	0.22	7.83

83.70	0.14	0.24	7.37
90.94	0.16	0.29	8.24
98.18	0.18	0.31	7.23
105.42	0.21	0.35	8.05
112.66	0.24	0.38	7.59
119.90	0.27	0.41	7.23
127.14	0.30	0.44	6.63
134.38	0.34	0.48	6.73
141.62	0.37	0.50	6.41
148.86	0.40	0.54	6.71
156.09	0.42	0.56	6.45
163.33	0.45	0.57	5.66

Dominance achieved at order 1.